

ANNAPAREDDY PAVAN KUMAR REDDY

DevOps Engineer | Site Reliability Engineer | AWS & Kubernetes

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PROFESSIONAL SUMMARY

DevOps/SRE Engineer with 4+ years of IT experience (3+ years hands-on in cloud infrastructure, CI/CD, and Kubernetes). Core experience at Tech Mahindra building CI/CD pipelines and managing AWS infrastructure; following role transition in early 2025, completed an intensive self-directed upskilling program building 4 production-grade DevOps projects on AWS EKS, including a self-healing GitOps platform with live auto-scaling validated under load testing. Proven ability to design, deploy, and troubleshoot Kubernetes workloads, Terraform-provisioned infrastructure, and GitOps pipelines end to end.

TECHNICAL SKILLS

- **Cloud:** AWS (EC2, S3, IAM, VPC, EKS, ECR, CloudWatch, ELB)
- **IaC:** Terraform (modular IaC, remote state, IAM roles)
- **Containers & Orchestration:** Docker, Docker Compose, Kubernetes (EKS), Helm
- **Scaling & GitOps:** HPA (Horizontal Pod Autoscaler), Metrics Server, ArgoCD (GitOps, Self-Heal)
- **CI/CD:** Jenkins, GitHub Actions
- **Monitoring & Logging:** Prometheus, Grafana, ELK Stack, CloudWatch
- **Configuration Management:** Ansible (playbooks, roles, handlers, variables)
- **Version Control:** Git, GitHub, GitOps workflows
- **Linux & Scripting:** Ubuntu, CentOS, Amazon Linux, Bash scripting
- **Programming:** Python (Flask), MySQL, YAML, JSON, Groovy
- **Security:** RBAC, IAM least-privilege policies, Security Groups, Kubernetes Secrets

PROFESSIONAL EXPERIENCE

DevOps Engineer | Tech Mahindra

Oct 2021 – Feb 2025

- Designed and maintained CI/CD pipelines using Jenkins and GitHub Actions, reducing deployment time by 40% and enabling automated build, test, and deployment workflows across multiple environments.
- Provisioned and managed AWS infrastructure (EC2, VPC, S3, IAM, EKS) using Terraform, achieving consistent and reproducible environment setups across dev, staging, and production.
- Containerized microservices using Docker and orchestrated deployments on Kubernetes (EKS), managing Pods, Deployments, Services, ConfigMaps, Secrets, Ingress, and HPA.
- Implemented GitOps-based continuous delivery using ArgoCD, enabling automatic synchronization of Kubernetes cluster state with GitHub repository configurations.
- Packaged and deployed applications using Helm charts, streamlining multi-environment deployments and enabling one-command rollbacks.
- Set up monitoring and alerting using Prometheus and Grafana, achieving 99.9% uptime visibility across services; integrated ELK Stack for centralized log management.
- Configured RBAC policies in Kubernetes to enforce least-privilege access for development teams across namespaces.
- Received Bravo Award for outstanding contribution to infrastructure automation and deployment reliability improvements.

IT Support Engineer | Megna Super IT Solutions

Jun 2020 – Oct 2021

- Provided L1/L2 technical support for Linux and Windows systems, resolving 95%+ of tickets within SLA timelines.
- Automated routine system administration tasks using Bash scripts, reducing manual effort by 30%.
- Managed user accounts, permissions, and system configurations across on-premise infrastructure.
- Monitored system health and proactively resolved incidents to minimize downtime.

KEY PROJECTS (Self-Directed, Post Tech Mahindra — Feb 2025 to Present)

Self-Healing GitOps Platform on AWS EKS — ArgoCD + HPA + Terraform

- Architected a self-healing, auto-scaling GitOps platform on AWS EKS where ArgoCD continuously syncs a 3-tier application (nginx, Flask, MySQL) from GitHub with zero manual kubectl intervention.
- Validated self-healing under chaos testing — manually deleted live pods and confirmed ArgoCD auto-detected drift and restored desired state within seconds.
- Implemented GitOps-managed Horizontal Pod Autoscaler (HPA) for frontend and backend tiers; load-tested with synthetic traffic and observed backend scale from 2 to 6 pods in real time as CPU utilization hit 99%.
- Diagnosed and resolved an EKS ENI-based pod-per-node scheduling limit on t3.small nodes during scale-out, right-sizing the node group via AWS CLI to restore cluster capacity under load.
- Resolved a Metrics Server TLS certificate verification failure blocking HPA metrics collection, applying a targeted kubelet configuration fix to restore live CPU/memory metrics cluster-wide.
- GitHub: github.com/Pavan-7673/blog-platform | github.com/Pavan-7673/terraform-eks

3-Tier Blog Platform — Flask + MySQL + Kubernetes + AWS EKS

- Designed and deployed a production-grade 3-tier application on AWS EKS: nginx frontend, Python Flask REST API backend (full CRUD), and MySQL 8.0 database, each in a separate Kubernetes Deployment within a dedicated namespace.
- Secured MySQL credentials using Kubernetes Secrets injected as environment variables, eliminating hardcoded credentials in application code; configured CPU/memory resource requests and limits across all 3 tiers.
- Exposed the frontend via a Kubernetes LoadBalancer Service, provisioning an AWS Elastic Load Balancer accessible via public DNS; internal tier communication routed through ClusterIP Services.

Ecommerce App on AWS EKS — Terraform + IAM + AWS Load Balancer

- Provisioned a production-grade AWS EKS cluster from scratch using modular Terraform code split across 7 files (provider.tf, variables.tf, iam.tf, eks.tf, etc.), following infrastructure-as-code best practices.
- Created dedicated least-privilege IAM roles for the EKS control plane and worker nodes (AmazonEKSClusterPolicy, AmazonEKS_CNI_Policy, AmazonEC2ContainerRegistryReadOnly) and pushed images to AWS ECR.
- Deployed an nginx-based frontend with 2 replicas behind an AWS Elastic Load Balancer, accessible via a public DNS URL with traffic distributed across pods.

Flask DevOps App — Docker + Helm + GitHub Actions CI/CD

- Containerized a Python Flask app with an optimized python:3.9-slim Dockerfile (60% smaller image) and packaged it as a Helm chart with parameterized values.yaml for dev/staging/prod deployments without code changes.
- Built a GitHub Actions CI pipeline that auto-builds and pushes Docker images to Docker Hub on every push, using GitHub Secrets for credential management; implemented Kubernetes RBAC for least-privilege namespace access.
- GitHub: github.com/Pavan-7673/devops-app-flask

EDUCATION

B.Tech in Computer Science Engineering

Acharya Nagarjuna University

CERTIFICATIONS & ACHIEVEMENTS

- Bravo Award — Tech Mahindra (2023) — Recognized for outstanding contribution to infrastructure automation
- Pat on the Back Award — Tech Mahindra — Recognized for proactive problem-solving and reliability contributions
- AWS Certified Cloud Practitioner — Exam scheduled